

Maximum speeds 435.115/117

Total axle ratio 6.53		Maximum speeds km/h at engine rated speed 2800/min																	
Main transmission	Main gears	Forwards								Reverse								Tires	
		1	2	3	4	5	6	7	8	1	2	3	4	5 ¹⁾	6 ¹⁾	7 ¹⁾	8 ¹⁾		
		6.0	8.7	13.2	17.9	26.4	38.1	57.7	78.5	6.2	9.0	13.6	18.5	27.2	39.2	59.4	80.9	18 - 20 12.5 - 20 ²⁾	
		6.3	9.1	13.7	18.6	27.5	39.7	60.1	81.7	6.5	9.3	14.1	19.3	28.4	40.8	61.9	84.2	12.5 R 20	
		6.3	9.1	13.8	18.8	27.8	40.0	60.6	82.5	6.5	9.4	14.3	19.4	28.6	41.2	62.5	85.0	14.5 - 20	
		6.6	9.5	14.4	19.6	28.9	41.7	63.2	85.9	6.8	9.8	14.9	20.3	29.8	43.0	65.1	88.6	14.5 R 20	
		6.1	8.7	13.2	18.0	26.5	38.2	57.9	78.8	6.2	9.0	13.6	18.6	27.3	39.4	59.6	81.2	9.00 - 20	
		6.1	8.9	13.4	18.2	26.9	38.8	58.7	79.9	6.3	9.1	13.8	18.8	27.7	39.9	60.5	82.4	9.00 R 20	

1) Locked with SA 35 985

2) standard

Total axle ratio 5.31		Maximum speeds km/h at engine rated speed 2800/min																	
Main transmission	Main gears	Forwards								Reverse								Tires	
		1	2	3	4	5	6	7	8	1	2	3	4	5 ¹⁾	6 ¹⁾	7 ¹⁾	8 ¹⁾		
		7.4	10.7	16.2	22.0	32.5	46.8	71.0	96.5	7.6	11.0	16.7	22.8	33.5	48.2	73.1	99.5	18 - 20 12.5 - 20 ²⁾	
		7.7	11.1	16.9	22.9	33.8	48.8	73.9	100.5	8.1	11.5	17.4	23.7	34.9	50.2	76.1	103.6	12.5 R 20	
		7.8	11.2	17.0	23.2	34.1	49.2	74.6	101.4	8.0	11.6	17.5	23.9	35.2	50.7	76.8	104.5	14.5 - 20	
		8.1	11.7	17.7	24.1	35.6	51.3	77.7	105.7	8.4	12.1	18.3	24.9	36.7	52.8	80.0	109.0	14.5 R 20	
		7.4	10.7	16.2	22.1	32.6	47.0	71.2	96.8	7.7	11.1	16.8	22.8	33.6	48.4	73.3	99.8	9.00 - 20	
		7.6	10.9	16.5	22.4	33.1	47.7	72.2	98.2	7.8	11.2	17.0	23.2	34.1	49.1	74.4	101.3	9.00 R 20	

1) Locked with SA 35 985

2) standard

Maximum speeds 435.110/.111/.113

Total axle ratio 6.38		Maximum speeds km/h at engine rated speed 2800/min																	
Main transmission	Main gears	Forwards								Reverse								Tires	
		1	2	3	4	5	6	7	8	1	2	3	4	5 ¹⁾	6 ¹⁾	7 ¹⁾	8 ¹⁾		
		6.5	9.4	14.2	19.3	28.4	41.0	62.1	84.4	6.7	9.7	14.6	19.9	29.3	42.2	63.9	87.0	14.5 - 20 16/70 - 24	
		6.8	9.8	14.8	20.1	29.6	42.7	64.7	88.0	7.0	10.1	15.2	20.7	30.5	44.0	66.6	90.7	12 R 22.5 ²⁾ 14.5 R 20	
		7.4	10.6	16.1	21.9	32.3	46.6	70.6	96.0	7.6	11.0	16.6	22.6	33.3	48.0	72.7	99.0	14.5 R 24	
		7.1	10.3	15.6	21.2	31.2	45.0	68.2	92.7	7.4	10.6	16.0	21.9	32.2	46.3	70.2	95.6	14.5 - 24	
		6.6	9.6	14.5	19.7	29.1	41.9	63.5	86.4	6.9	9.9	14.9	20.4	30.0	43.2	65.4	89.0	11.00 - 20 11.00 R 20	
		7.0	10.1	15.2	20.7	30.6	44.1	66.8	90.8	7.2	10.4	15.7	21.4	31.5	45.4	68.8	93.7	12.00 R 20 13 R 22.5 16/70 - 24 14.75/80 R 20	
		6.0	9.9	15.0	20.4	30.1	43.3	65.6	89.3	7.1	10.2	15.4	21.1	31.0	44.6	67.6	92.0	12.00 - 20	

1) Locked with SA 35 985

2) standard

Total axle ratio 5.40		Maximum speeds km/h at engine rated speed 2800/min																	
Main transmission	Main gears	Forwards								Reverse								Tires	
		1	2	3	4	5	6	7	8	1	2	3	4	5 ¹⁾	6 ¹⁾	7 ¹⁾	8 ¹⁾		
		7.66	11.05	16.72	22.75	33.55	48.39	73.31	99.64	7.91	11.40	17.24	23.50	34.50	49.82	75.47	102.69	14.5 - 20 16/70 - 24	
		7.98	11.52	17.44	23.73	34.99	50.47	76.47	103.94	8.25	11.89	17.98	24.51	36.06	51.97	78.72	107.11	12.5 R 22.5 ²⁾ 14.5 R 20	
		8.71	12.56	19.02	25.87	38.15	55.03	83.37	113.32	8.99	12.96	19.61	26.72	39.35	56.66	85.82	116.78	14.5 R 24	
		8.41	12.14	18.38	25.00	36.87	53.18	80.57	109.51	8.69	12.53	18.95	25.82	38.03	54.76	82.94	112.85	14.5 - 24	
		7.84	11.31	17.11	23.79	34.33	49.53	75.03	101.99	8.09	11.67	17.65	24.05	35.42	50.99	77.24	105.10	11.00 - 20 11.00 R 20	
		8.25	11.90	18.01	24.51	36.13	52.12	78.96	107.33	8.52	12.28	18.57	25.31	37.27	53.67	81.29	110.61	12.00 R 20 13 R 22.5 16/70 - 24 14.75/80 R 20	
		8.11	16.69	17.70	24.08	35.51	51.22	77.59	105.47	8.37	12.07	18.25	24.87	36.63	52.73	79.88	108.69	12.00 - 20	

1) Locked with SA 35 985

2) standard

Maximum speeds 435.110/111 (transmission model 717.901)

Total axle ratio 6,38		Maximum speeds km/h at engine rated speed 2800/min																
Main transmission	Main gears	Forwards								Reverse								Tires
		1	2	3	4	5	6	7	8	1	2	3	4	5 ¹⁾	6 ¹⁾	7 ¹⁾	8 ¹⁾	
		6,5	9,4	14,2	19,3	28,4	41,0	62,1	84,4	6,7	9,7	14,6	19,9	29,3	42,2	63,9	87,0	14,5 - 20 16/70 - 24
		6,8	9,8	14,8	20,1	29,6	42,7	64,7	88,0	7,0	10,1	15,2	20,7	30,5	44,0	66,6	90,7	12 R 22,5 ³⁾ 14,5 R 20
		7,4	10,6	16,1	21,9	32,3	46,6	70,6	96,0	7,6	11,0	16,6	22,6	33,3	48,0	72,7	99,0	14,5 R 24
		7,1	10,3	15,6	21,2	31,2	45,0	68,2	92,7	7,4	10,6	16,0	21,9	32,2	46,3	70,2	95,6	14,5 - 24
		6,6	9,6	14,5	19,7	29,1	41,9	63,5	86,4	6,9	9,9	14,9	20,4	30,0	43,2	65,4	89,0	11,00 - 20 11,00 R 20
		7,0	10,1	15,2	20,7	30,6	44,1	66,8	90,8	7,2	10,4	15,7	21,4	31,5	45,4	68,8	93,7	12,00 R 20 13 R 22,5 14,75/80 R 20 ²⁾
		6,9	9,9	15,0	20,4	30,1	43,3	65,6	89,3	7,1	10,2	15,4	21,1	31,0	44,6	67,6	92,0	12,00 - 20
Additional gear box	working gears	1,13	1,62	2,46	3,35	4,93	7,11	10,77	14,65	1,16	1,68	2,54	3,46	5,09	7,33	11,10	15,11	14,5 - 20 16/70 - 24
		1,17	1,69	2,56	3,49	5,14	7,41	11,23	15,27	1,21	1,75	2,64	3,60	5,30	7,64	11,57	15,74	12 R 22,5 ³⁾ 14,5 R 20
		1,28	1,85	2,80	3,80	5,61	8,09	12,25	16,66	1,32	1,91	2,88	3,93	5,79	8,33	12,62	17,18	14,5 R 24
		1,24	1,78	2,70	3,67	5,42	7,81	11,83	16,09	1,28	1,84	2,78	3,80	5,59	8,05	12,19	16,59	14,5 - 24
		1,15	1,66	2,62	3,42	5,05	7,28	11,03	15,00	1,19	1,72	2,59	3,54	5,21	7,50	11,36	15,46	11,00 - 20 11,00 R 20
		1,21	1,75	2,65	3,60	5,31	7,66	11,60	15,77	1,25	1,81	2,73	3,72	5,48	7,89	11,95	16,26	12,00 R 20 13 R 22,5 14,75/80 R 20 ²⁾
		1,19	1,72	2,60	3,54	5,22	7,52	11,40	15,50	1,23	1,77	2,68	3,66	5,38	7,75	11,74	15,98	12,00 - 20
	crawler gears	0,116	0,167	0,253	0,345	0,509	0,733	1,111	1,511	0,120	0,173	0,261	0,356	0,524	0,755	1,144	1,557	14,5 - 20 16/70 - 24
		0,121	0,175	0,264	0,359	0,530	0,764	1,157	1,574	0,125	0,180	0,272	0,371	0,547	0,787	1,193	1,623	12 R 22,5 ³⁾ 14,5 R 20
		0,132	0,190	0,288	0,392	0,578	0,834	1,263	1,718	0,136	0,197	0,297	0,405	0,597	0,859	1,301	1,771	14,5 R 24
		0,128	0,184	0,278	0,379	0,559	0,805	1,220	1,659	0,132	0,190	0,287	0,391	0,576	0,830	1,257	1,710	14,5 - 24
		0,119	0,171	0,259	0,353	0,521	0,750	1,137	1,564	0,123	0,177	0,267	0,365	0,537	0,773	1,171	1,594	11,00 - 20 11,00 R 20
		0,125	0,180	0,273	0,371	0,547	0,789	1,196	1,626	0,129	0,186	0,281	0,383	0,585	0,813	1,232	1,676	12,00 R 20 13 R 22,5 14,75/80 R 20 ²⁾
		0,123	0,177	0,268	0,365	0,538	0,776	1,175	1,598	0,127	0,183	0,276	0,377	0,555	0,799	1,210	1,647	12,00 - 20

1) Locked with SA 35 985

2) 13.00 - 20 Pilote XL

3) Standard

Maximum speeds 435.110/.111 (transmission model 717.900)

Total axle ratio 6,38		Maximum speeds km/h at engine rated speed 2800/min																
Main transmission	Main gears	Forwards								Reverse								Tires
		1	2	3	4	5	6	7	8	1	2	3	4	5 ¹⁾	6 ¹⁾	7 ¹⁾	8 ¹⁾	
		9,0	12,5	17,1	23,3	32,7	45,4	62,1	84,4	9,3	13,0	17,7	24,1	33,8	46,9	64,0	87,0	
		9,4	13,1	17,9	24,3	34,1	47,3	64,7	88,0	9,7	13,5	18,4	25,1	35,2	48,9	66,7	90,7	
		10,3	14,3	19,5	26,5	37,2	51,6	70,6	96,0	10,6	14,7	20,1	27,4	38,4	53,4	72,8	99,0	
		9,9	13,8	18,8	25,6	35,9	49,8	68,2	92,7	10,2	14,2	19,4	26,4	37,1	51,5	70,3	95,6	
		9,2	12,8	17,5	23,9	33,4	46,4	63,5	86,4	9,5	13,3	18,1	24,6	34,6	48,0	65,5	89,1	
		9,7	13,5	18,4	25,1	35,2	48,9	66,8	90,9	10,0	13,9	19,0	25,9	36,4	50,5	68,9	93,7	
		9,6	13,3	18,1	24,7	34,6	48,0	65,6	89,3	9,9	13,7	18,7	25,4	35,7	49,6	67,7	92,1	
Additional gear box	working gears	1,57	2,18	2,97	4,05	5,68	7,88	10,78	14,66	1,62	2,25	3,07	4,18	5,86	8,14	11,11	15,11	14,5 - 20 16/70 - 24
		1,63	2,27	3,10	4,22	5,92	8,21	11,23	15,28	1,69	2,34	3,20	4,35	6,11	8,49	11,57	15,75	12 R 22,5 ³⁾ 14,5 R 20
		1,78	2,48	3,38	4,61	6,46	8,96	12,26	16,67	1,84	2,56	3,49	4,75	6,67	9,26	12,63	17,19	14,5 R 24
		1,72	2,39	3,27	4,45	6,24	8,66	11,84	16,10	1,78	2,47	3,37	4,59	6,44	8,94	12,20	16,60	14,5 - 24
		1,60	2,23	3,04	4,14	5,81	8,07	11,03	15,00	1,66	2,30	3,14	4,27	6,00	8,33	11,37	15,47	11,00 - 20 11,00 R 20
		1,69	2,34	3,20	4,36	6,12	8,48	11,60	15,78	1,74	2,42	3,30	4,50	6,31	8,77	11,95	16,27	12,00 R 20 13 R 22,5 14,75/80 R 20 ²⁾
		1,66	2,30	3,15	4,28	6,01	8,34	11,40	15,51	1,71	2,38	3,24	4,42	6,20	8,61	11,75	15,98	12,00 - 20
	crawler gears	0,162	0,225	0,307	0,418	0,586	0,813	1,111	1,511	0,167	0,232	0,316	0,431	0,605	0,840	1,145	1,558	14,5 - 20 16/70 - 24
		0,168	0,234	0,319	0,435	0,610	0,847	1,158	1,575	0,174	0,242	0,330	0,449	0,630	0,875	1,193	1,624	12 R 22,5 ³⁾ 14,5 R 20
		0,184	0,255	0,349	0,475	0,666	0,924	1,264	1,719	0,190	0,264	0,360	0,490	0,688	0,955	1,302	1,772	14,5 R 24
		0,178	0,247	0,337	0,459	0,643	0,892	1,221	1,660	0,183	0,255	0,347	0,473	0,664	0,922	1,258	1,711	14,5 - 24
		0,165	0,230	0,314	0,427	0,600	0,832	1,137	1,547	0,171	0,237	0,324	0,441	0,619	0,859	1,172	1,595	11,00 - 20 11,00 R 20
		0,174	0,242	0,330	0,449	0,631	0,875	1,196	1,627	0,180	0,250	0,340	0,463	0,651	0,904	1,232	1,677	12,00 R 20 13 R 22,5 14,75/80 R 20 ²⁾
		0,171	0,238	0,324	0,442	0,620	0,860	1,175	1,600	0,176	0,245	0,334	0,455	0,693	0,888	1,211	1,648	12,00 - 20

1) Locked with SA 35 985

2) 13.00 - 20 Pilote XL

3) Standard

Weights and Trailer Loads
Weights

Chassis			Tires			Axle loads		Weights Unladen weight			
Model	Sales designation	Version	Size	Type	PR	FA kg	RA kg	Chas- sis kg	with platform kg	Permissible payload kg	Total weight kg
435.115 .117	U 1300 L U 1300 L/ 37	Series	12,5 – 20	MPT	10	3700	3700	3750	5252 ¹⁾	1 ²⁾	7400
			14,5 – 20	MPT	10	3960	3960				7490 ³⁾
			14,5 R 20	MPT	10						
			12,5 – 20	MPT	12	4000	4000				
			12,5 R 20	MPT	12						
14,5 R 20		14									
9,0 – 20		14									
			9,0 R 20		14						
.115	U 1300 L Special vehicle with box-type body		12,5 – 20	MPT	12				6130		7490

1) Varies depending on equipment

2) Corresponding to specifications of body manufacturer

Chassis m			Tires			Axle loads		Weights Unladen weight						
Model	Sales designation	Version	Size	Type	PR	FA	RA	Chas- sis	with platform	Permissible payload	Permis- sible total weight kg			
						kg	kg	kg	kg	kg				
435.111	U 1700 L	Series	11,00 – 20		16	5300	5300	4450	5600 ¹⁾	¹⁾²⁾	9000			
			11,00 R 20		16									
			12,00 – 20		18									
			12,00 R 20		18									
			12 R 22,5			5300	5300							
			13 R 22,5											
		14,5 – 20	MPT	18	5290							5290		
		14,5 R 20	MPT	18										
		14,5 – 24	MPT	16									5300	5300
		14,5 R 24	MPT	16										
16/70 – 24	MPT	14												
14,75/80 R 20/			5300	5800										
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18			5290	5290								
14,5 R 20	MPT	18												
14,5 – 24	MPT	16	5300	5400										
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT	14					5300	5800						
11,00 – 20		16							5300	5800				
11,00 R 20		16												
12,00 – 20		18												
12,00 R 20		18												
14,75/80 R 20/														
13,00 – 20 – Pilot XL														
12 R 22,5														
13 R 22,5														
14,5 – 20	MPT	18	5290	5290										
14,5 R 20	MPT	18												
14,5 – 24	MPT	16			5300	5400								
14,5 R 24	MPT	16												
16/70 – 24	MPT													

1) Varies depending on equipment

2) For tractor unit - chassis, determined during acceptance

3) Special vehicle

4) 5600 kg with SA 35 968, 35 969, 35 779, 35 810

5) Max. 4000 kg for cross-country applications

MPT = Multipurpose tire

Loads

Chassis			Tires			Axle loads		Weights			
Model	Sailes designation	Version	Size	Type	PR	FA	RA	Unladen weights Chassis	with platform	perm. payload	Total weight
						kg	kg	kg	kg	kg	kg
435.110	U 1700	Series ²⁾	14,5 R 20	MPT	18						
			14,5 - 20	MPT	18						
			14,5 R 24	MPT	16						
			14,5 - 24	MPT	16						
			16/70 - 24	MPT	14						
			11,00 R 20		16	5300	5300	4450	5820	3180	9000
			11,00 - 20		16						
			12 R 22,5								
			13 R 22,5								
			12,00 R 20		18						
			12,00 - 20		18						
			14,75/80 R 20								
			13,00 - 20 - Pilote XL								
		Series ²⁾ with SA 35810	14,5 R 20	MPT	18						
			14,5 - 20	MPT	18						
			14,5 R 24	MPT	16						
			14,5 - 24	MPT	16						
			16/70 - 24	MPT	14						
			11,00 R 20		16	5300	5300	4450	5820	3440	10000
			11,00 - 20		16						
			12 R 22,5								
			13 R 22,5								
			12,00 R 20		18						
			12,00 - 20		18						
			14,75/80 R 20								
			13,00 - 20 - Pilote XL								
		with ³⁾ SA 35810	14,5 R 20	MPT	18	5300	5450			3600	
			14,5 - 20	MPT	18						
		SA 35969	14,5 R 24	MPT	16						
			14,5 - 24	MPT	16						
			16/70 - 24	MPT	14	5300	6140	4450	5820		
			11,00 R 20		16						
			11,00 - 20		16						
			12 R 22,5			5300	6200			4240	10600
			13 R 22,5								
			12,00 R 20		18						
			12,00 - 20		18						
			14,75/80 R 20								
			13,00 - 20 Pilote XL								
		with ³⁾ SA 35776	14,5 R 20	MPT	18	5450	5450			3600	
			14,5 - 20	MPT	18						
		SA 35810	14,5 R 24	MPT	16						
			14,5 - 24	MPT	16						
			16/70 - 24	MPT	14	5600	6140				
			11,00 R 20		16			4450	5820		10600
			11,00 - 20		16						
		SA 35968	12 R 22,5			5600	6200			4240	
			13 R 22,5								
		SA 35969	12,00 R 20		18						
			12,00 - 20		18						
			14,75/80 R 20								
			13,00 - 20 Pilote XL								

- 1) applicable only for tractive unit and equal weight distribution on the tractive unit chassis.
Decided when body is agreed.

- 2) and as tractive unit and tractive unit chassis
3) tractive unit and tractive unit chassis

MPT = Multipurpose tire

Authorized towed weights**Trailers with brakes**

Chassis			Trailer, gross weight	
Model	Sales designation	Gross weight of towing vehicle kg	with non-continuous brake (over-run brake) kg	with continuous brake kg
435.115	U 1300 L	7 490	7020 to 7500	10 480
435.111	U 1700 L	9 000 10 000 10 600	8 000	12 600 14 000 14 840

Trailers without brakes

Chassis			Trailer
Model	Sales designation	Gross weight of towing vehicle kg	Gross weight kg
435.115	U 1300 L	7 490	2 300
435.111	U 1700 L	9 000 10 000 10 600	2 000

Trailers with brakes

Chassis ⁴⁾			Trailer, gross weight		Trailer coupling ¹⁾
Model	Sales designation	Gross weight of towing vehicle kg	with non-continuous brake (over-run brake) kg	with continuous brake kg	
435.110	U 1700	9 000	8 000	29 000	all
		10 000 ³⁾		28 000	
		10 600 ^{1) 3)}		25 600	
		10 600 ^{2) 3)}		27 400	64 G 135 or 605
.113	U 1700 L/38	12 000	—	12 000	225 G 110 64 G 135

1) see technical data 31.3 - 1.1/1

Trailers without brakes

Chassis ⁴⁾			Trailer
Model	Sales designation	Gross weight of towing vehicle kg	Gross weight kg
435.110	U 1700	9 000	2 000
		10 000 ³⁾	
		10 600 ^{1) 3)}	
		10 600 ^{2) 3)}	
435.113	U 1700 L/38	12 000	2 500

1) with reinforced rear axle

2) with reinforced rear- and front axle and steering LS 7 F, SA 35 968, SA 35 969, SA 35 776

3) exhaust brake SA 35 810

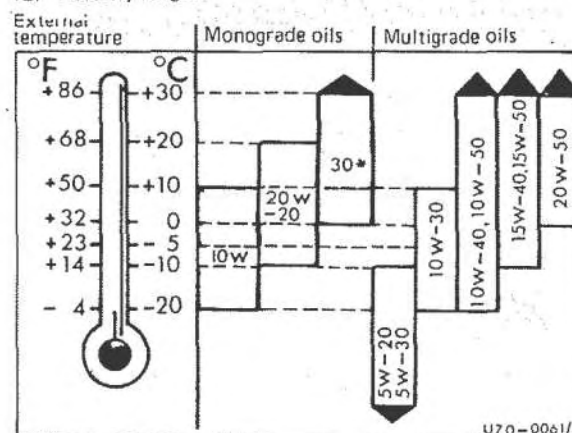
4) tractor and chassis tractor

Filling capacities

Assembly		Service product (page No. of MB specification for service products	SAE class	Season	Capacity (l) ¹⁾ 435.115 .117 435.110 .111 .113	
Engine	with oil filter	Engine oil (226-227.1)	12)	12)	max.15	max.15
	proportion in oil filter				min. 12	min. 12
					2 x 1.0	
					1.5	2 x 1.0
Lock-up torque converter		Engine oil (226-227.1)	10 W 7)	all year round	17.5	
Clutch operation	Control reservoir	Brake fluid ⁵⁾ (331.1 331.2)	—	all year round	0.2	
Transmission	Main transmission	Engine oil ¹⁾ ⁶⁾ (226-227.1)	10 W	all year r.	10.5	
	- with low ratio gears	Transmission oil (235.1)	80 ⁹⁾ 80 W ⁹⁾ 80 W/ 85 W ⁹⁾ 80W/90 ¹⁰⁾ 90 ¹⁰⁾	all year round	11.5	
	- with crawler gears		12.5			
	- with pto 1:1		9.5			
	- with pto 0.7 : 1		10.0			
Hydrostat		Engine oil	10 W ³⁾	all year round	—	17.0
		Hydraulic oil				
Pto shaft drive	540/1000/min	Transmission oil (235.1)	80	all year round	5.75	
		Engine oil ¹⁾ (226-227.1)	10			
	3 700/min	Transmission oil (235.1)	80		8.0	
Pto shafts	Shaft bearings ⁸⁾	Transmission oil (235.1)	80	all year round	0.1	0.1
		Engine oil ¹⁾ (226-227.1)	10 W			
Axles	Central housing	Hypoid gear oil (235)	90	all year r.	2.5	2.5
	Wheel hub drive				ea.0.3	ea.0.3
	Differential lock	Initial operation oil (225)	10 W		0.001	—
Steering	Hydraulic steering	Engine oil (226, 227)	10 W ³⁾	all year round		
		or ATF (236.2)	—		2.25	2.25
		Hydraulic oil (226,227,236,2)				3.25 ⁴⁾
Braking circuit	Complete capacity	Brake fluid ⁵⁾ (331.1)	—	all year r.	app.0.8	app.1.0
Compressed air brakes	with anti-freeze bottle	Ethyl alcohol (methylated spirits)	—	all year round	app.0.2	app.0.2
		Iso propyl alcohol ¹⁾				
Grease nipples	Axle, chassis, cab	Lubricating grease (266)	—	all year r.	as required	
Winch	at front	Transmission oil	80	all year r.	2.0	2.0

- 1) Applicable for military vehicles
a) Only those oils specified by Daimler-Benz may be used.
b) Use oils specified by the vehicle owner. If he has not left any instructions, use the normal oils specified.
2) Listed oils may not be mixed with each other.
3) Use SAE 5 W-20/30 in cold regions.
4) Applicable for LS 7.
5) Change yearly (safety requirement).
6) Only applicable for transmission model 717.901 without auxiliary transmission.
7) In cold regions SAE 5 W-20/30
In hot regions SAE 30
8) Long-term filling
9) Optional
10) In hot zones
11) With turbocharged engines use only S3 grade engine oil

12) Viscosity ranges



* With a continuous external temperature above +30°C (86°F) SAE 40 can be used.

Assembly		Service product (page No. of MB specification for service products)	Season	Capacity (l)	
				435.115 .117	435.110 .111 .113
Battery	- Terminals	Acid-resistant grease FT 40 VI (350) tech. vaseline ¹⁾	all year round	as required	
	- Topping up	Distilled water			
Start pilot	Supply reservoir	Starter fuel	all year r.	0.05	0.05
Fuel system	Fuel tank	Diesel fuel DIN 51 601	all year r.	160	160
Cooling circuit	with heater	Coolant ⁷⁾ (325)	all year round	20	20
	proportion of anti-freeze down to -25°C -13°F (standard filling)	Anti-freeze (325)		8	8
	proportion of anti-freeze down to -40°C -40°F	Anti-freeze (325)		10.2	10.2
	and corrosion inhibitor	Corrosion inhibitor (311)		0.2	0.2
Screen wash circuit	Screen wash bottle	MB screen wash additive	all year round	9.5	7

1) Applicable for military vehicles

- a) Only those oils specified by Daimler-Benz may be used.
- b) Use oils specified by the vehicle owner. If he has not left any instructions, use the normal oils specified.

7) Consisting of water, anti-freeze and corrosion inhibitor.

Special versions U 1300 L considered:

Engine:	16.457/26	18.638/03	39.975/01	83.036/01
	35.737/04	35.738/02	35.949/02	35.977/01
Clutch:	35.741			
Transmission:	35.737/04	35.738/02	35.949/02	35.977/01
	35.985/01			
Frame:	35.014/17	35.149/04	35.238/03	35.460/09
	35.780/01	35.817/08	35.851/02	35.951/01
	35.951/02	35.976/01		
Suspension:	35.633/02	35.772/04		
Axle:	35.973/01	35.974/01		
Brake:	35.734/10	35.822/01	35.823/02	35.966/01
	35.981/01	35.988/01	35.992/01	35.997/01
	35.998/01			
Steering:	35.987/01			
Fuel system:	35.778/02	35.921/01		
Electrical equipment:	35.496/05	35.647/03	35.775/03	35.769/05
	35.769/10	35.769/11	35.827/01	35.827/02
	35.828/01	35.831/01	35.912/05	35.942/01
	35.965/02	35.979/01	35.980/01	35.986/01
	35.993/01	53.482/01	82.509/54	82.983/24
Cab:	35.765/10	35.928/09	53.472/01	53.473/02
	53.474/03	53.477/02	53.478/02	53.479/03
	53.480/03	53.483/07	53.484/02	53.485/02
	53.486/01	53.489/02	53.491/06	53.492/01
	53.495/01	53.497/01	53.950/01	53.951/01
	53.962/01	53.977/01	53.977/02	53.978/01
	53.980/01	53.982/02	53.983/01	
Platform:	435.611	35.017/10	35.074/03	35.149/04
	35.270/09	35.270/14	35.534/07	35.761/01
	35.762/02	35.763/02	35.764/01	35.766/02
	35.767/01	35.768/01	35.777/01	35.825/01
	35.982/01	70.071/01		
Special parts:	35.847/02	35.951/02		

If a particular unit incorporates only one or two special versions (SA's), these will be indicated beneath the heading.

The job descriptions may differ for vehicles which do not feature corresponding special versions (SA's).

Special versions U 1700 L/38 considered:

Engine:	16.074	18.631	18.638	35.810
	35.975	81.408	82.773	82.928
	83.017	83.036	83.069	
Clutch:	35.741			
Transmission:	35.737	35.738	35.749	35.970
	35.977	35.985		
Frame:	35.872	35.999		
Suspension:	35.301	35.772		
Axle:	36.140			
Wheels:	36.103			
Brake:	35.643	35.810	35.966	35.976
	36.043	36.089		
Steering:	35.776			
Fuel system:	35.270	35.778	35.921	
Electrical equipment:	35.098	35.647	35.693	35.703
	35.769	35.827	35.965	35.979
	35.980	35.986	35.992	35.993
	36.017	36.089	36.139	53.482
	53.484	53.496	53.957	53.985
	82.509	82.983		
Cab:	35.847	36.048	53.472	53.473
	53.474	53.475	53.480	53.482
	53.483	53.484	53.485	53.489
	53.491	53.493	53.499	53.950
	53.951	53.955	53.958	53.978
	53.980	53.982	53.984	53.985
	53.991			
Platform:	35.074	35.270	35.437	35.460
	35.503	35.534	35.559	35.761
	35.762	35.763	35.764	35.768
	35.777	35.825	36.105	36.133
	36.137	39.725	39.729	

Where only one or two special versions (SA) of a unit are fitted, this/these is/are listed in the heading.

The description of jobs may differ in the case of vehicles without corresponding special versions (SA).

Installation Survey

Chassis Model	Sales designation	Series Chassis end No.	Engine Model	Sales designation	Series/SA Engine end No.
435.110	U 1700	from 002 230	353.959	352 A. V/2 352 A. LII/1 ⁵⁾	Series from 577 967
.111	U 1700 L ¹⁾	from 000 008			Series from 413 968
.113	U 1700 L/38	from 084 642			Series from 693 577
.115	U 1300 L ¹⁾	from 000 019	353.961	352.X/2	Series from 482 095
		from 000 061 ²⁾		352.X/1 ²⁾	SA 16 457/26 ²⁾ SA 35 975 ²⁾ SA 83 036 ²⁾ from 509 465
		from PB ³⁾	353.959	352 A. V/2 352 A. LII/1 ⁵⁾	SA 36 019 ⁴⁾ from PB ³⁾
.117	U 1300 L/37	from PB ³⁾	353.961	352.X/2	Series from PB ³⁾
		from 071 796	353.959	352 A. V/2 352 A. LII/1 ⁵⁾	SA 36 019 ⁴⁾ from 627 976
.160	Attachment TM 170	from PB ³⁾	353.959	352 A. LII/1	from PB ³⁾
.170	Attachment Condor	from PB ³⁾	353.977	352 A. V/2	Series from 575 178

1) L = Truck

2) Special vehicle BW

3) PB = Start of production

4) Valid only for fire service vehicles

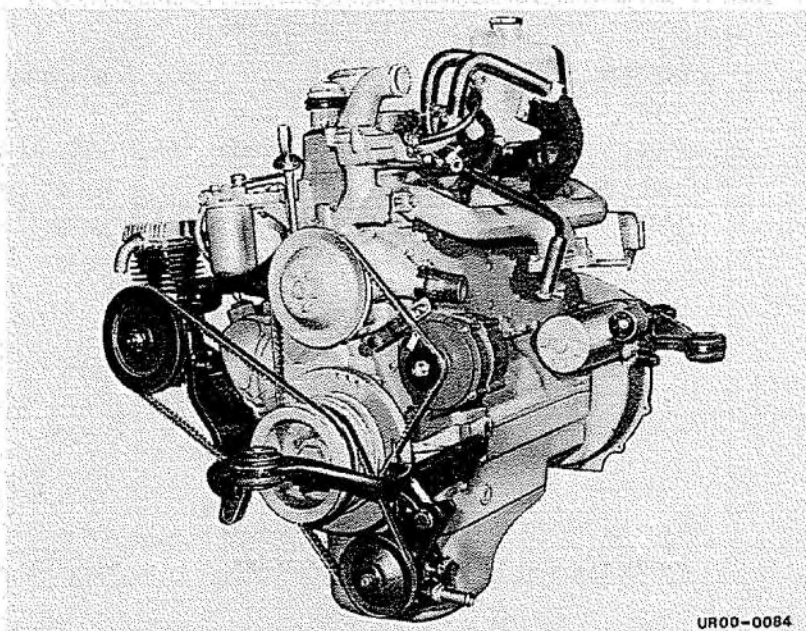
5) From engine end No. 691 150

MBMANUALS.COM

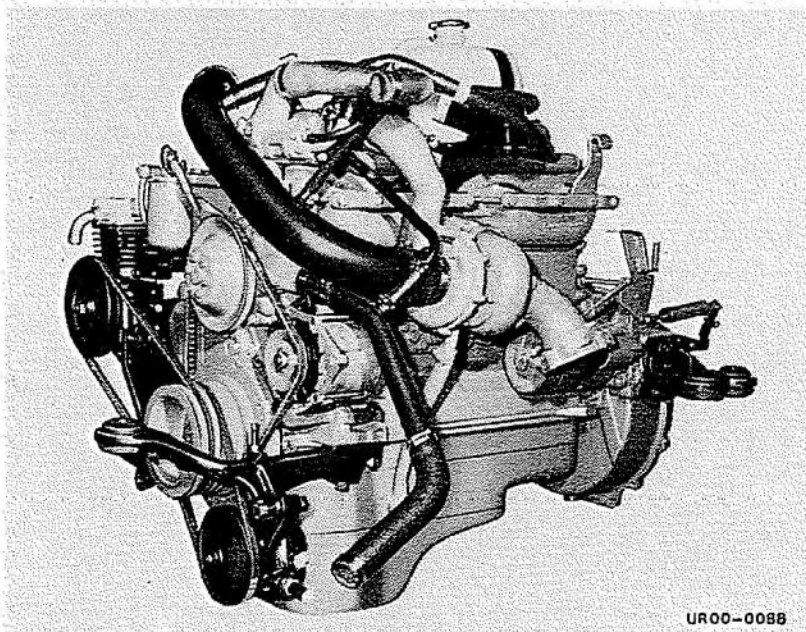
01.8 General

352/352 A

General View

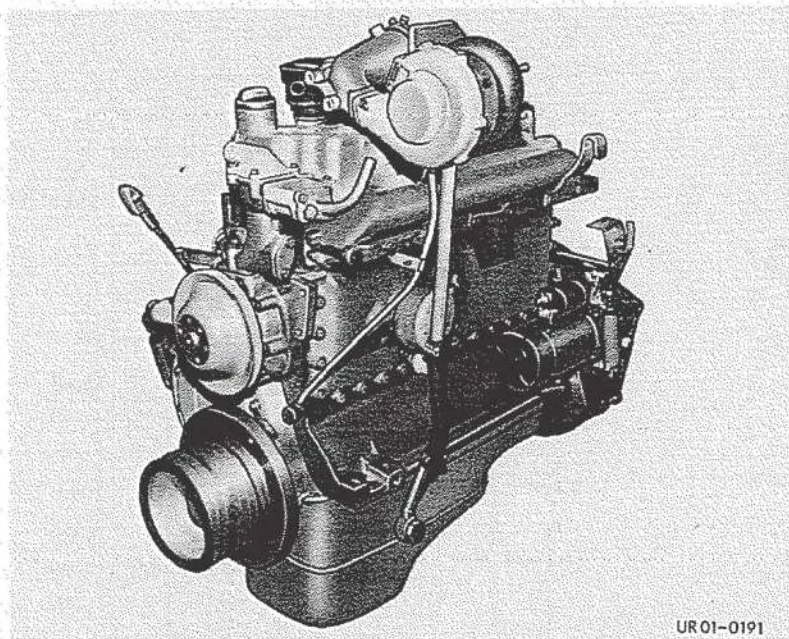


Engine 353.961



Engine 353.959

General view

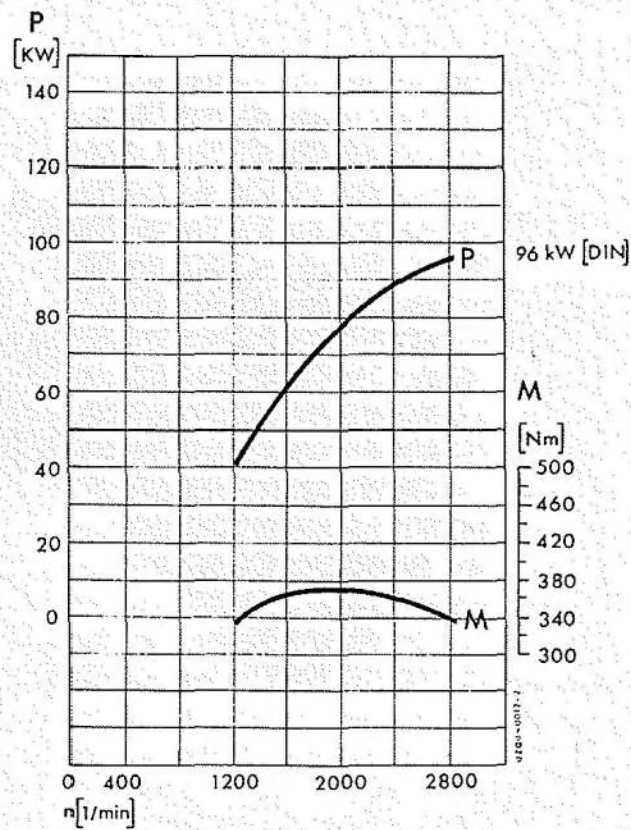


UR 01-0191

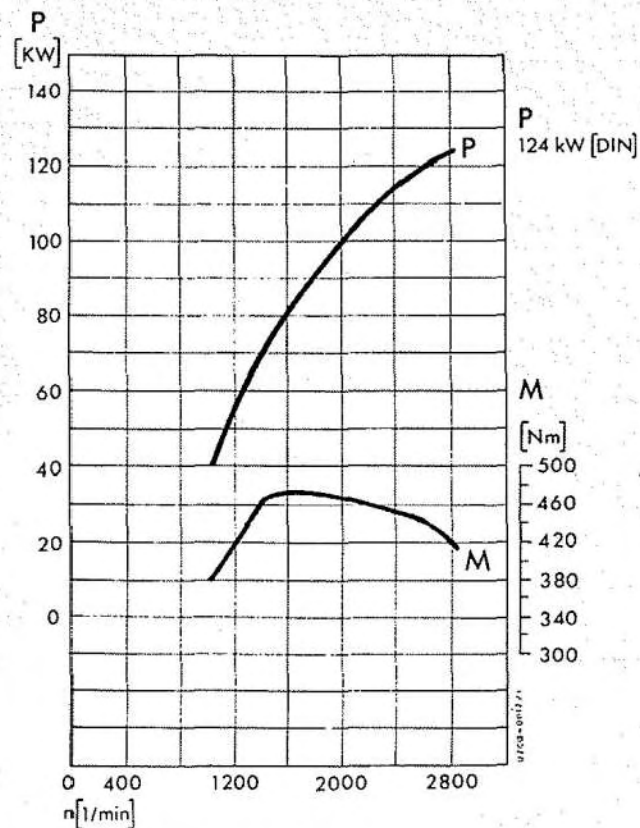
Engine 353.977

MBMANUALS.COM

Performance diagram

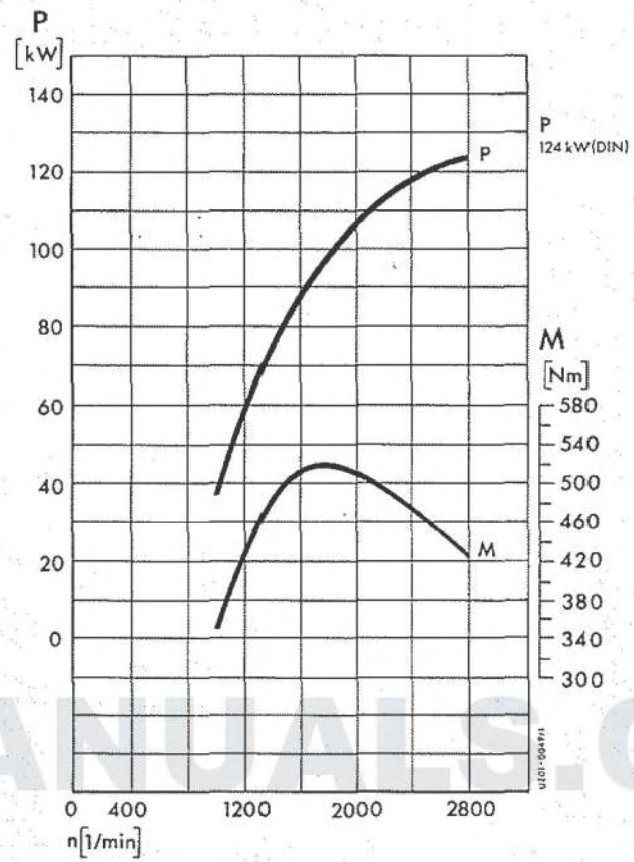


Engine 353.961



Engine 353.957 to engine end No. 691 149
.977

Performance Diagram



Engine 353.959 from engine end No. 691 150

Technical Data

Mercedes Benz type			352	352 A		
Model			353.961	353.959	353.977	
Combustion system			Four-stroke diesel with direct injection	Four-stroke diesel with direct injection		
Number of cylinders Cylinder arrangement			6, vertical in-line			
Bore ø	mm		97			
Stroke	mm		128			
Total displacement	cm ³		5657			
Compression ratio			17	16		
Compression at starting speed		bar	22 up to 24 min. 20			
Continuous output at nominal speed		kW (HP) 1/min	96 (130) 2800	124 (168) 2800		
Maximum torque at speed		Nm 1/min	363 1700	491 ¹⁾ 1600	530 ²⁾ 1600	491 1600
Idle speed		1/min	700			
Start of delivery before TDC		°	18	21 ¹⁾	19 ²⁾	21
Injection sequence			1 - 5 - 3 - 6 - 2 - 4			
Valve clearance, cold	Inlet	mm	0,20		0,25 ²⁾	0,20
	Exhaust	mm	0,30		0,40 ²⁾	0,30
Opening pressure	new	bar	200 + 10			
of injection nozzles	used	bar	min. 180			
Operating temperature		° C	95			
Oil pressure	normal	bar	2 to 5			
	at idle speed	bar	min. 0.6			
Weight, dry		kg	460		480	

1) To end engine No. 691 149.

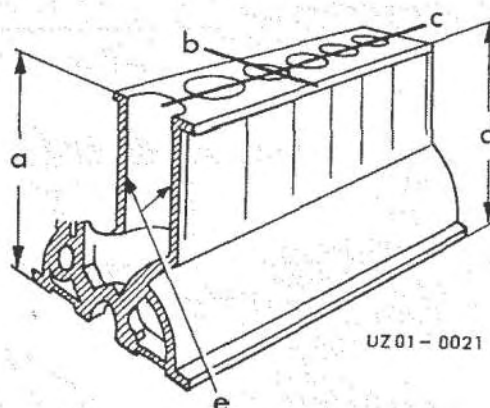
2) From end engine No. 691 150.

01.8 General

352/352 A

Crankcase

Engine Type			352	352 A
Overall height of crankcase (Production size)	a	mm	<u>359,10</u> 359,00	
Repair stage I ¹⁾		mm	<u>358,80</u> 358,70	
Repair stage II ¹⁾		mm	<u>358,50</u> 358,40	
Repair stage III ¹⁾		mm	<u>358,20</u> 358,10	
Permissible unevenness of mating surface	Crosswise	b	mm	0,017
	Lengthwise	c	mm	0,03
Peak-to-valley height of upper mating surface		mm	0,008 up to 0,016	
Permissible parallel misalignment of upper to lower mating surface in lengthwise direction	d	mm	0,1	
Pressure test in water		bar	2,5	
Piston may recede	= -	mm	0,07	
Piston may project	= +	mm	0,30	



1) If mating surface is remachined to next repair stage, pistons of reduced compression height must be installed.

Cylinder Bores (e)

Normal mm	Normal I mm	Normal II mm	Repair stage I mm	Repair stage II mm	Repair stage III mm
97,010 96,990	97,085 97,065	97,135 97,115	97,510 ¹⁾ 97,490	98,010 ¹⁾ 97,990	98,510 ¹⁾ 98,490
97,010 ²⁾ 96,990	-	-	97,010 ²⁾ 96,990		
Permissible ovality			mm	0,01	
Permissible conicity			mm	0,01	
Permissible deviation of cylinder bores perpendicular to crankshaft axis, relative to 200 mm length			mm	0,04	
Peak-to-valley height of the cylinder bores			mm	0,003 up to 0,005	
Waviness of cylinder bores			mm	max. 50% of peak-to-valley height	
Wear in liner at top reversal point of 1st piston ring			mm	max. 0,12	
Wear in liner in centre of liner			mm	max. 0,05	
Bore depth for machining to the repair stages			mm	min. 250	

1) Not engine 353.959/ 977.

2) Possible repair- crankcase with dry cylinder liners may be installed. If necessary, the scope of work can be requested.

Camshaft Bearings

Basic bores in crankcase	mm	60,030 60,000
--------------------------	----	------------------

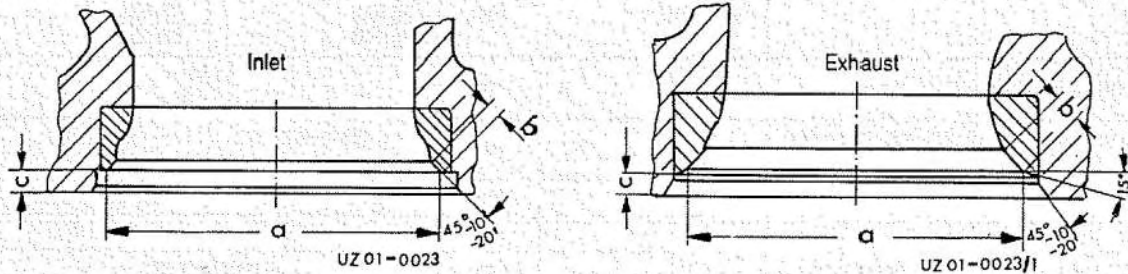
Valve tappet bores

Normal mm	Repair stage I mm	Repair stage II mm	Repair stage III mm
28,033 28,000	28,233 28,200	28,533 28,500	28,783 28,750

Cylinder head

Overall height of cylinder head	mm	92,10 91,90
Permissible material allowance	mm	1,00
Permissible roughness of mating surface	lengthwise	mm 0,05
	crosswise	mm 0,015
Permissible parallel misalignment of upper to lower mating surface	lengthwise	mm 0,10
Pressure test in water	bar	2,5
Nozzle projection relative to lower edge of cylinder head	mm	1.8 to 2.5
Peak-to-valley height of lower mating surface	mm	0,008 up to 0,016

Valve seat



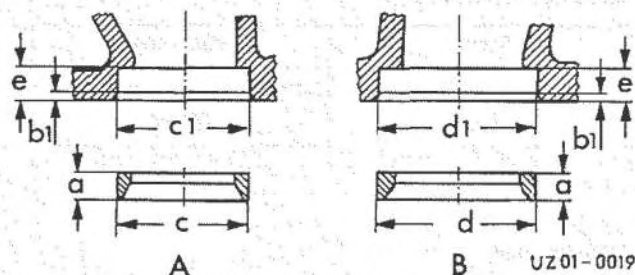
Valve seat ring diameter for new rings	a	Inlet	mm	42 ± 0,01
		Exhaust	mm	34 ± 0,01
Width	b	Inlet	mm	1,6 up to 2,0
		Exhaust	mm	2,0 up to 2,5
Reference dimension	c	Inlet	mm	2,8 + 0,2
		Exhaust	mm	2,6 + 0,2
Permissible runout of valve seat			mm	0,03
Distance between cylinder head mating surface and valve head		Inlet	mm	0,7 + 0,5
		Exhaust	mm	0,7 + 0,5
Valve seat angle		Inlet and exhaust	°	45° - 10' - 20'
Correction cut	Inside	Inlet	°	60°
		Exhaust	°	75°
	Outside	Inlet and exhaust	°	30°

Valve seat rings must be renewed if the machining limit is exceeded when remachining the valve seat.

01.8 General

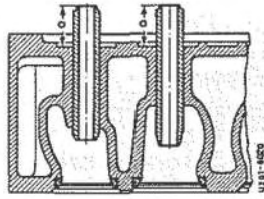
352/352 A

Valve Seat Rings



		Exhaust A		Inlet B	
		1)		1)	
Outside diameter valve seat ring c and d	Normal	mm	38,080 38,070	45,080 45,070	45,880 45,870
	Repair stage I	mm	38,380 38,370	45,380 45,370	46,180 46,170
	Repair stage II	mm	38,580 38,570	45,580 45,570	—
Bore diameter in cylinder head c 1 and d 1	Normal	mm	38,025 38,000	45,025 45,000	45,825 45,800
	Repair stage I	mm	38,325 38,300	45,325 45,300	46,125 46,100
	Repair stage II	mm	38,525 38,500	45,525 45,500	—
Installation depth in cylinder head	e	mm	11,20 11,10		
Distance between cylinder head face and valve seat ring	b 1	mm	2,6 + 0,2 2,8 + 0,2		
Basic bore in cylinder head		mm	38,000 + 0,025 45,000 + 0,025		
Height of valve seat rings	a	mm	8,5 - 0,1 8,3 - 0,1		
Valve seat ring overlap in cylinder head		mm	0,045 to 0,080		

Valve Guides

Distance between valve guide valve spring seat	a	mm	Exhaust 24 - 0,5	Inlet 24 - 0,5	
Inside diameter		mm	10,022 10,000	9,022 9,000	
Length		mm	73	78	
	Normal	mm	Repair stage I	Repair stage II	Repair stage III
Outside diameter	15,046 15,028		15,146 15,128	15,246 15,228	15,546 15,528
Bore in cylinder head	15,018 15,000		15,118 15,100	15,218 15,200	15,518 15,500
Overlap	mm	+ 0,010 to + 0,046			

1) Engine model 353.959 from engine end No. 549 882.

Special Tools

Cons. No.	Tool	Tool No.	Tool Set
1	Ring wrench for injection lines	000 589 07 03 00	A
2	Pliers for connecting nipple of rocker shafts	312 589 01 37 00	A
3	Claw wrench for protective sleeve in cylinder head	346 589 00 07 00	B
4	30 mm socket for pressure screw in nozzle holder	000 589 75 09 00	B
5	Extractor for nozzle holder	352 589 00 33 00	B
6	Centering drift	425 589 00 61 00	B
7	Drift for removing inlet valve guide	110 589 02 15 00	C
8	Drift for removing exhaust valve guide	615 589 01 15 00	C
9	Fitting drift for valve seat rings	346 589 03 15 00	C
10	Installer for valve guide	352 589 00 43 00	C
11	Reamer, 9 mm diameter for inlet valve guide	000 589 10 53 00	C
12	Reamer, 10 mm diameter for exhaust valve guide	000 589 11 53 00	C
13	Reamer for valve guide sleeve bore in cylinder head	000 589 18 53 00	C
14	Hand cutter for valve seat rings	000 589 12 66 00	C
15	Special wrench to retighten injection lines	000 589 68 03 00	D
16	Limit plug gauge dia. 9 mm inlet valve guide	116 589 08 21 00	D
17	Connection piece for compression test	352 589 00 21 00	D
18	Limit plug gauge, 10 mm dia. for exhaust valve guide	617 589 05 23 00	D
19	Handle for cylinder head (2 necessary)	312 589 01 31 00	D
20	Rud special lifting chane	435 589 00 62 00	D

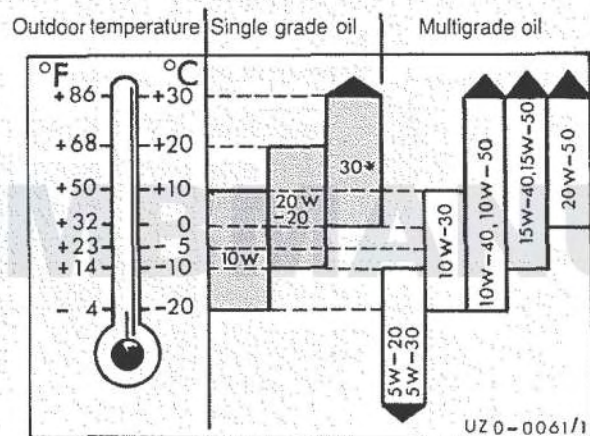
01.8 General

352/352 A

Capacities

Assembly			Service product (page No. of MB service product specification)	SAE-grade	Capacities (l)	
					352 ³⁾	352 A
Engine	With oil filter	max.	Engine oil (224,226)	1)	14,5	15,0
		min.			11,5	12,0
	Proportion in oil filter		(227.0, 227.1) ⁵⁾		1,5	2 x 1,0
Cooling system	With heater		Coolant ⁴⁾ (325)		20	
	Antifreeze proportion down to -25°C -13°F		Antifreeze (325)		8	
	(Standart filling)					
	Antifreeze proportion down to -40°C -40°F		Antifreeze (325)		10,2	
And corrosion protection			Anticorrosion oil (311)		0,2	

1) Viscosity grades



*) SAE 40 may be used if outdoor temperatures are permanently above +30°C (86°F).

2) To engine end No. 502 247

3) From engine end No. 502 248

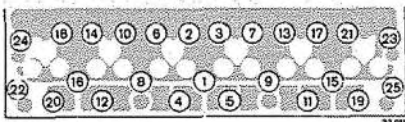
4) Consisting of water, antifreeze and anticorrosion oil

5) For supercharged engines, only use engine oil of "S 3" quality.

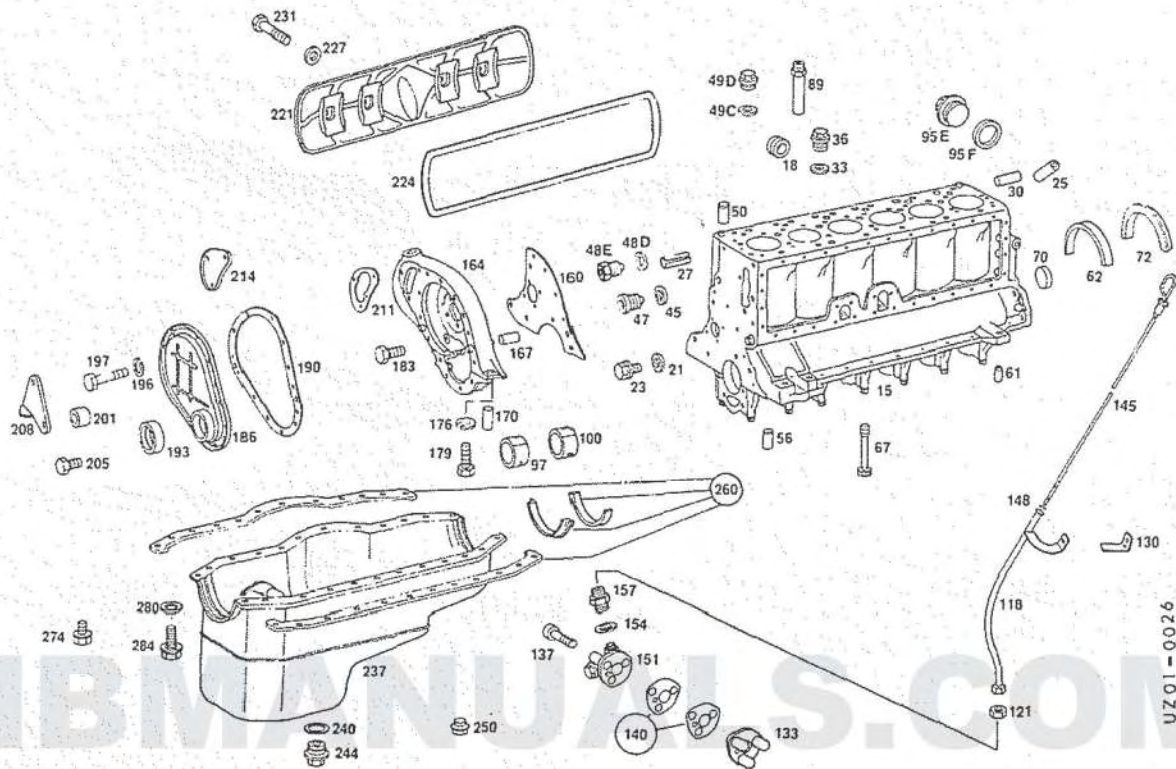
Necessary material

Cons. No.	Designation	Part No.
1	Loctite Nr. 241	002 989 70 71
2	Loctite Nr. 573	001 989 45 20
3	Curil T	001 989 37 20
4	Curil K 2	commercially available
5	Loctite Nr. 221	002 989 69 71

Tightening Torques

Designation	Thread	Nm	Torquing angle
Cylinder head cover	M 8	25	
Push rod cover to crankcase	M8	4 to 6	
Timing gear case cover	Casting	M 6	8
	Sheetmetal	M 6	8
Cylinder head (tightened in 3 stages according to tightening diagram)			
	1st step	M 12	60
	2nd step	M 12	90
	3rd step	M 12	110
Oil pan (sheet steel)	M 6	8	
	M 8	9	
Oil drain plug to oil pan	M 26	80 + 10	
Protective sleeve for nozzle holder in cylinder head	M 14	60	
Nozzle holder in cylinder head, thrust screw	M 34	60 to 70	
Tapered plug in cylinder head	M 30	100 to 200	
	M 24	70 to 150	
Coolant line on top of cylinder head	M 8	25	
Flywheel housing to crankcase	M 12	80	
Timing gear case to crankcase	M 10	65	
Screw plug for oil passage in crankcase	M 20	100 ± 10	
Screw plug for oil duct, front and rear	M 16	40 + 5	
Flywheel to crankshaft	Initial torque	M 12	30 + 10
	Final torque	M 12	— 90° + 20°
Rocker arm brackets assembly to cylinder head	M 12	100 up to 110	
Leakage fuel line to injection nozzle	M 8	15 up to 20	
Union nut of injection line	to injection pump	M 12	25
	to nozzle holder	M 14	25
Air compressor bracket to engine mount		75	
Exhaust manifold to cylinder head	M 10	30	
Exhaust manifold to cylinder head, 2-hole metal plate	M 10	25	
Flanged manifold to exhaust pipe	M 10	45	
Front engine mounting to frame	M 14	140 ± 20	
Rear engine mounting to frame	M 14	140 ± 20	
Front engine support to crankcase	M 14	170 ± 20	
Left engine support arm to flywheel housing	M 12	110 ± 20	

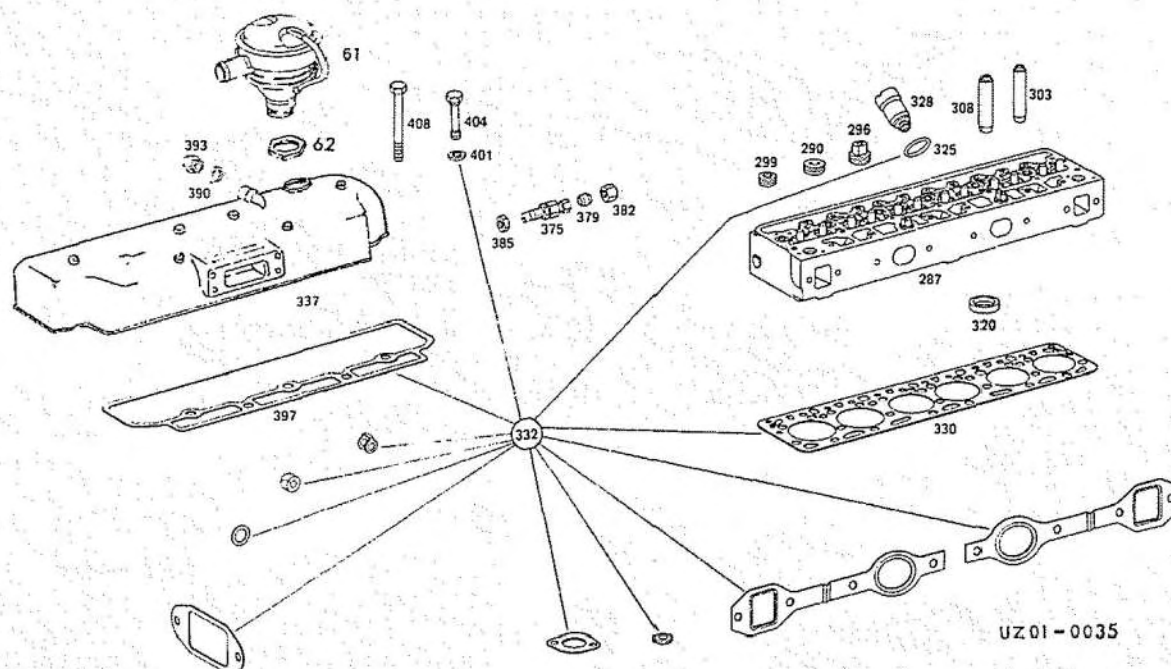
Exploded view



Engine housing

15	Crankcase	95E	Screw plug	186	Timing gear case cover
18	Screw plug	95F	Sealing ring	190	Gasket
21	Sealing ring	97	Camshaft bearing No. 1	193	Shaft seal
23	Screw plug	100	Camshaft bearing Nos. 2,3,4	196	Lock washer
25	Sleeve	118	Guide tube	197	Bolt
27	Sleeve	121	Cap nut	201	Spacing ring
30	Straight pin	130	Bracket	205	Bolt
33	Sealing ring	133	Flange	208	Pointer
36	Screw plug	137	Bolt	211	Gasket
45	Sealing ring	140	Gasket	214	Cover
45	Screw plug	145	Oil dipstick	221	Pushrod cover
48D	Sealing ring	148	Sealing ring	224	Gasket
48E	Screw plug	151	Flange	227	Sealing ring
49C	Sealing ring	154	Sealing ring	231	Bolt
49D	Screw plug	157	Union	237	Oil pan
56	Straight pin	160	Gasket	240	Sealing ring
61	Straight pin	164	Timing gear case	244	Screw plug
62	Fabric sealing ring	167	Straight pin	260	Repair set
67	Bolt	170	Straight pin	274	Bolt
70	Cover	176	Lock washer	280	Washer
72	Gasket	179	Bolt	284	Bolt
89	Guide tube	183	Bolt		

Exploded view



Cylinder head

61	Engine breather filter	320	Valve seat ring
62	Nut	325	Sealing ring
287	Cylinder head	328	Protective bushing
290	Screw plug	330	Cylinder head gasket
296	Stopper	332	Gasket set
299	Screw plug	337	Cylinder head cover
303	Intake valve guide	339	End cover
308	Exhaust valve guide	342	Sealing ring

Cylinder bore diagnosis



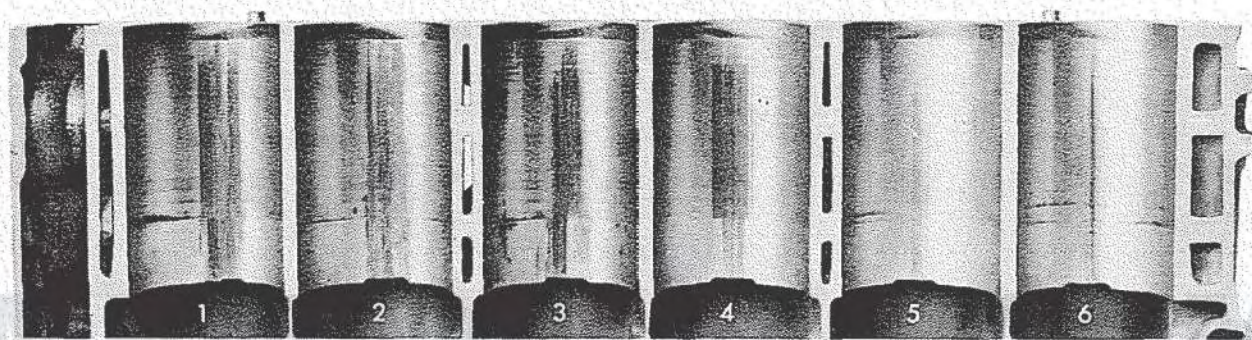
Owner

2. Branch/agents

3. Date

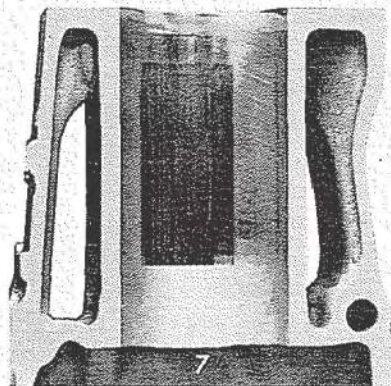
4. Repair job No.

5. Place	6. Registration No.	7. Date of 1st registration	8. Model	9. Chassis No.	10. Mileage
11. Copy	12. Presented by	13. Time accepted	14. Accepted by	15. Engine No.	16. Provisional deadline

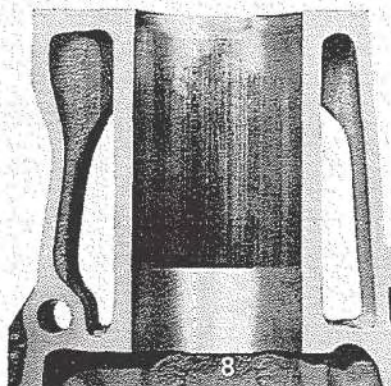


R-3088

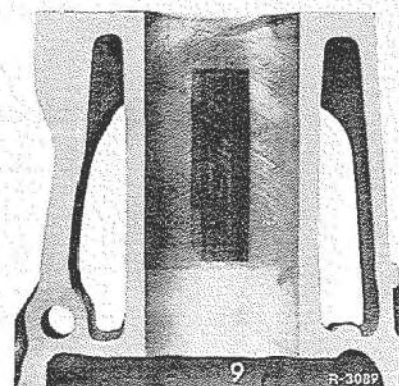
- Cyl. 1-3 Piston skirt has seized. Honing pattern destroyed. Engine has to be removed.
 Cyl. 4 Piston skirt and oil scraper ring have seized. Honing pattern destroyed. Engine has to be removed.
 Cyl. 5-6 Piston skirt cracked. Honing pattern still complete. Engine can continue working.



Ring has seized at land side, approx. 30 mm wide. Step due to oil ring is quite evident. Honing pattern destroyed. Engine has to be removed.



Oil ring seizure has extended across entire cylinder half. Honing pattern destroyed. Engine has to be removed.



Semi-severe oil ring seizure. Honing pattern is still quite evident. Engine can continue working.



Contents

Chapter	Page
1 General	
Installation survey	1.1/1
Section drawing	1.1/2
Performance diagram	1.1/4
Technical data	1.2/1
Special tools	1.3/1
Filling capacities	1.3/2
Necessary materials	1.3/2
Tightening torques	1.4/1
Exploded view	1.5/1
Cylinder bores diagnosis	1.6/1
2 Reconditioning engine	
Removal and installation of engine	2.1/1
Removal and installation of engine parts	2.3/1
3 Removal and installation of cylinder head cover	3.1/1
4 Removal and installation of cylinder head	4.1/1
5 Disassembly and assembly of cylinder head	5.1/1
6 Reconditioning of cylinder head	
Checking, exchanging valve guides	6.1/1
Refacing valve seats with valve seat cutter	6.2/1
Refacing valve seats with hand cutter	6.3/1
Removal and installation of valve seat rings	6.4/1
Remachining basic bore for valve seat rings	6.5/1
7 Checking compression pressures	7.1/1

Removal

Note: Engine removal is similar for both engines No. 353.959 and 353.961.

- 1 Raise driver's cab, see 60.5-2.1/1 or 60.5-2.2/1.
- 2 Disconnect battery.

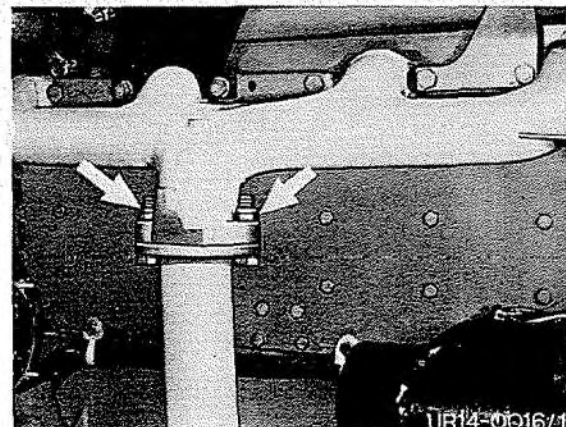
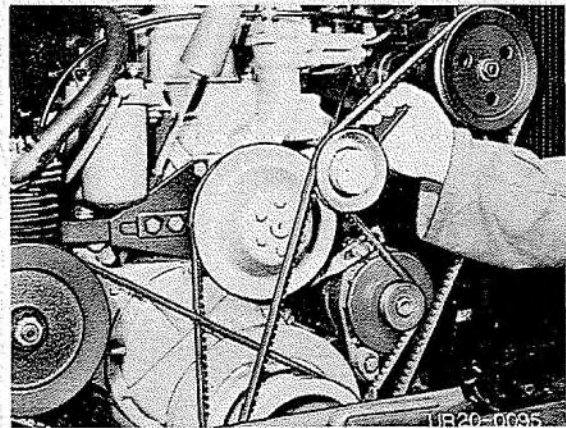
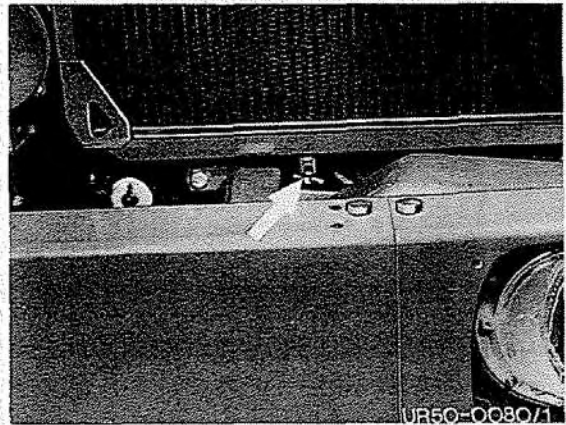
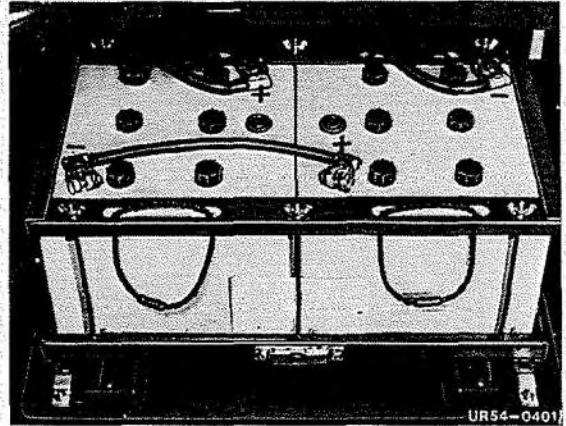
Note: Firstly disconnect (-) then (+) terminal.

- 3 Drain coolant.

- 4 Remove V-belt from intermediate fan shaft.

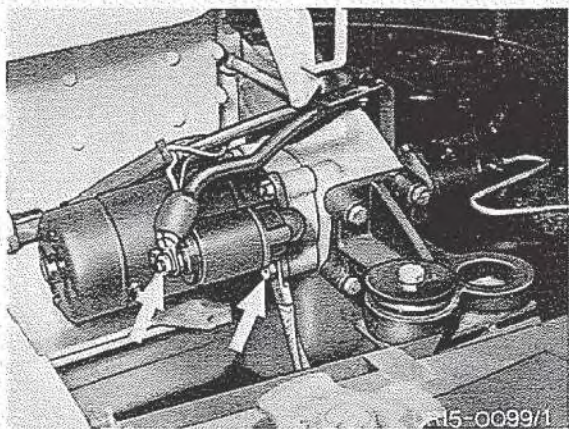
- 5 Detach exhaust pipe from exhaust manifold.

Note: For engine model 353.959, detach exhaust pipe at turbocharger.

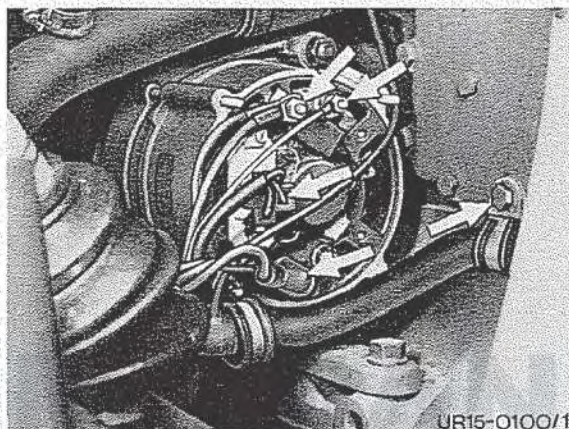


01.8 Removal and Installation of Engine

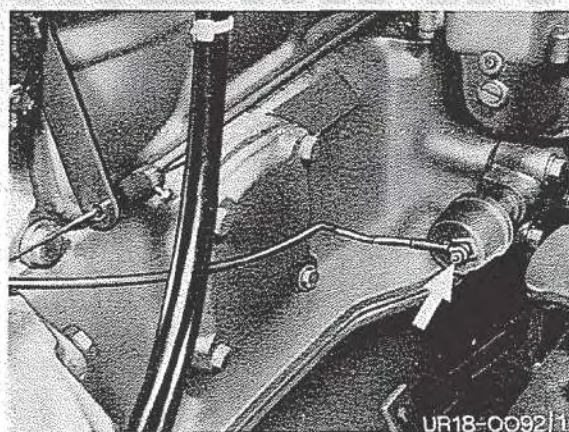
352 / 352 A



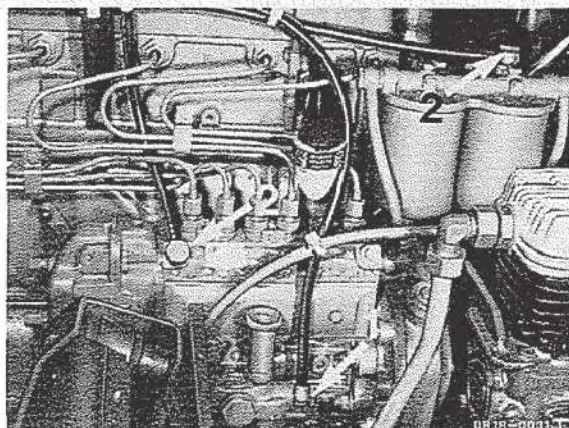
6 Disconnect electrical lines and ground strap from starter.



7 Mark and detach alternator wiring harness at oil cooler and electrical lines at the alternator.



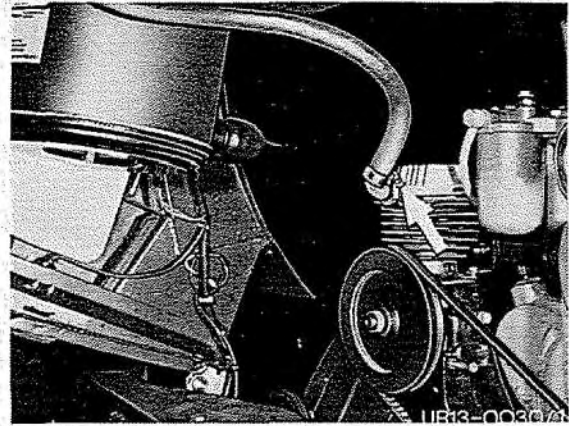
8 Detach electrical line at oil pressure pickup and coolant temperature pickup.



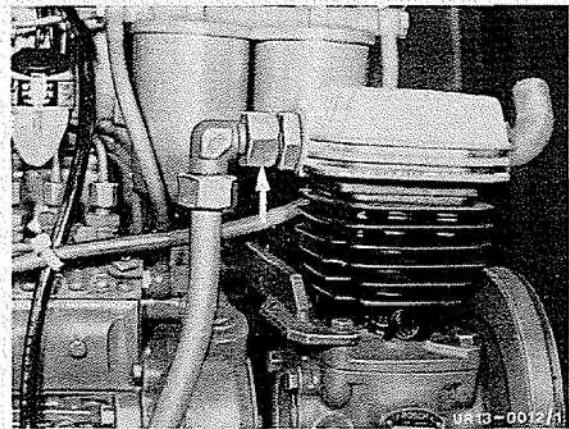
9 Remove fuel intake line (1) and return lines (2).

Note: The bleeding facility on the fuel filter is no longer fitted from chassis end No. 086 284.

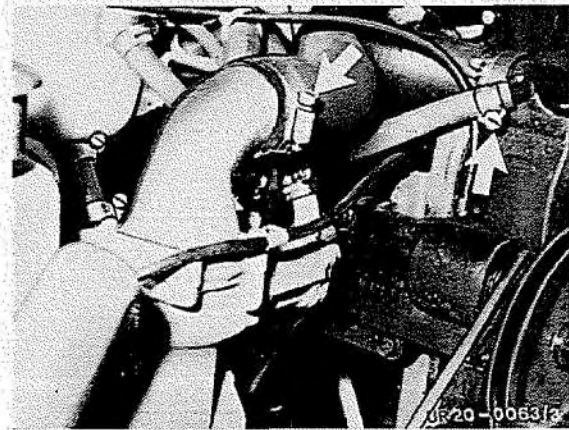
10 Detach intake line at air compressor.



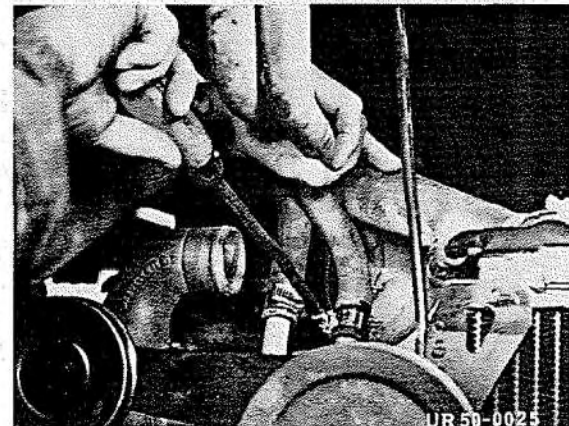
11 Detach delivery line at air compressor.

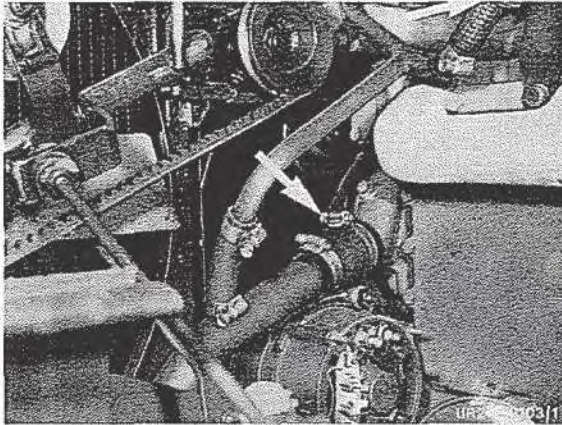


12 Detach coolant hose at thermostat body and coolant venting hose at radiator.

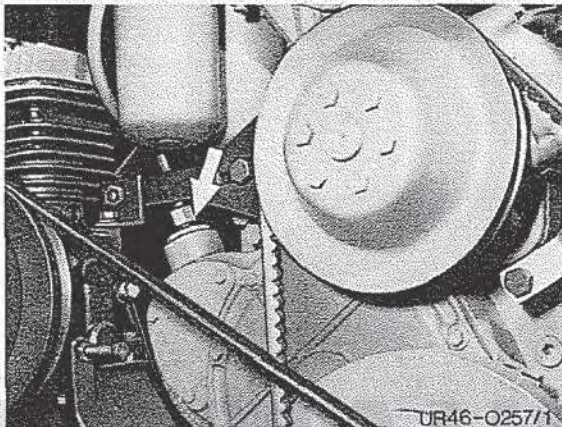


13 Detach heater hoses.





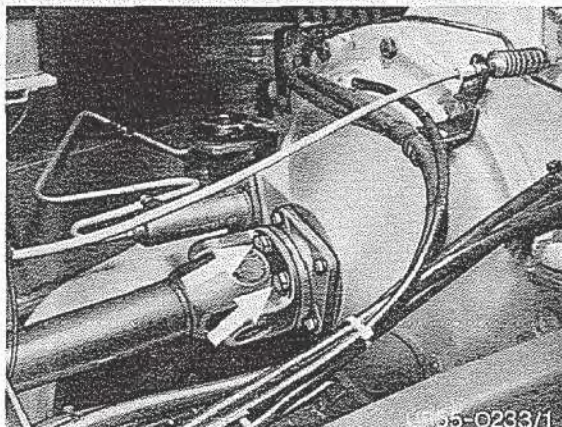
- 14 Detach coolant pipe at coolant pump and reservoir.



- 15 Detach venting line between timing gear case and steering fluid reservoir.

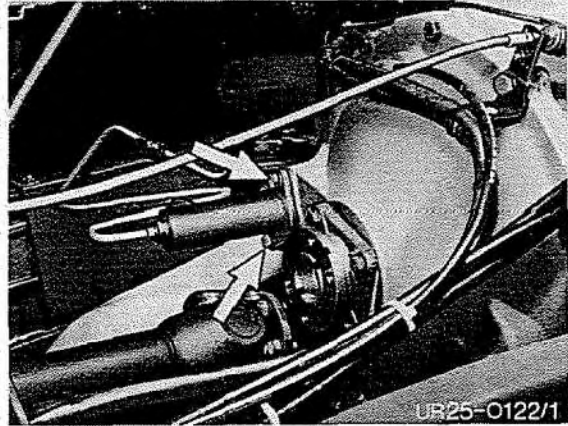


- 16 Detach hydraulic pump and bracket from engine support. Do not remove lines.

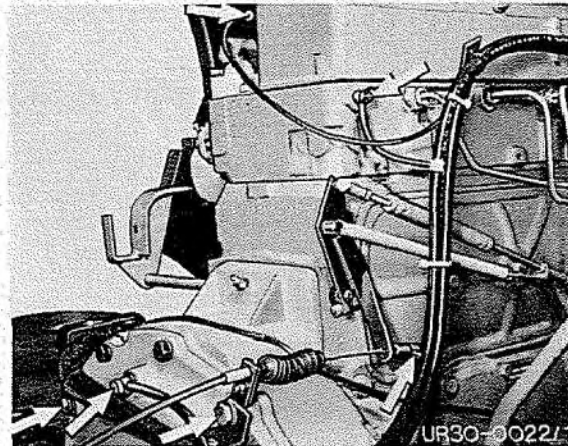


- 17 Detach propeller shaft from input flange.

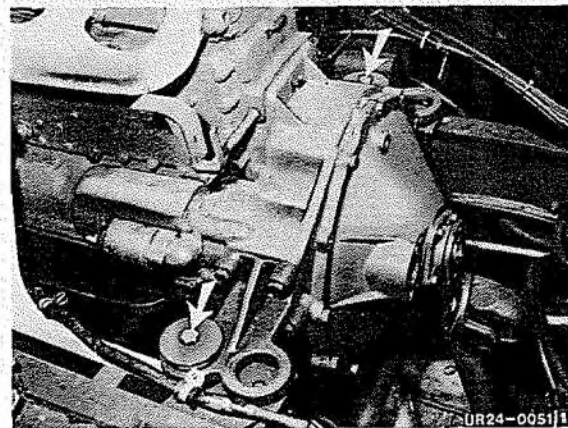
18 Remove clutch slave cylinder, do not detach line.



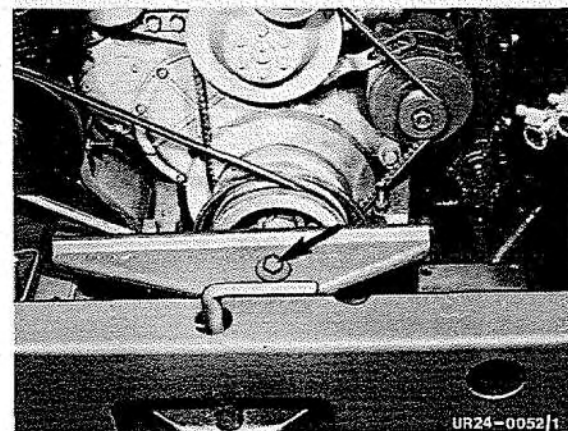
19 Detach starter wiring harness and venting line from clutch housing. Also remove fuel line, leakage oil line, cold starting line and control cable for idling if SA 35 989 installed.



20 Remove fastening bolts for rear engine mounts.

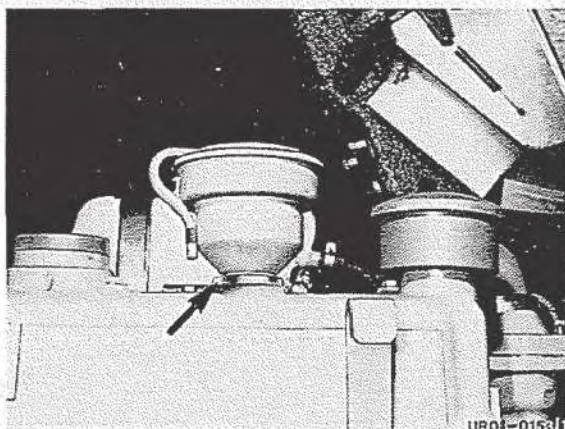


21 Remove fastening bolt at front engine mount. Detach cover.



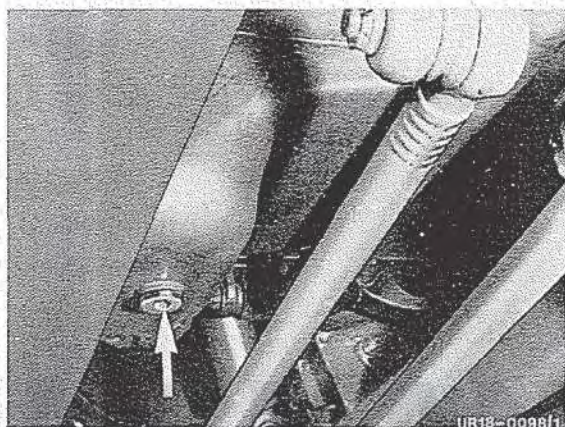
01.8 Removal and Installation of Engine

352 / 352 A

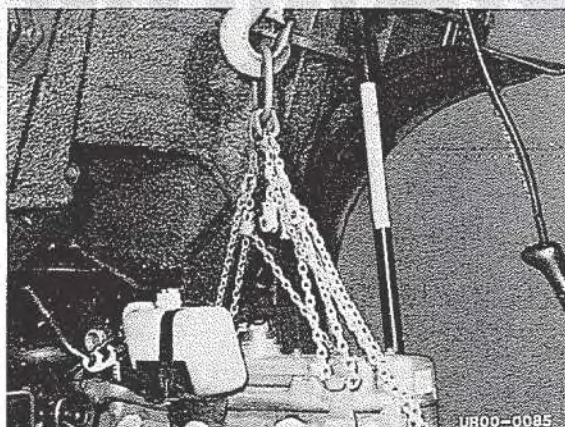


22 Detach engine breather filter.

Note: Renew engine breather filter without the code marking "E" every two years. A damaged engine breather filter must always be renewed immediately.



23 Drain engine oil as required.



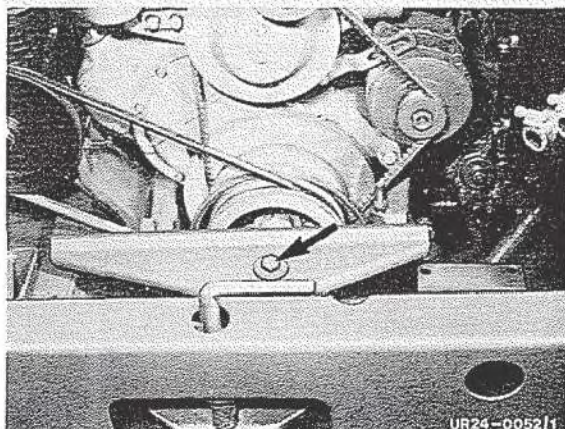
24 Lift out engine using special tool No. 19.

25 Clean all parts, check and replace if necessary.

Installation

Note: Engine installation is similar for both engine models No. 353.959 and 353.961.

1 Slide engine into position using special tool No. 19.



2 Fit cover and fastening bolt to front engine mount and tighten bolt.
Tightening torques, see 1.4/1.